Theorem 2. *`Pre-Norm`: Semantic subspaces must be represented as orthogonal spheres $\mathbb{S}^{N_\alpha} \equiv \mathcal{S}^{N_\alpha - 1}$ defined using the L_2-norm. Corollary: if either orthogonality or constant-norm are violated, semantic subspaces interfere through a multiplicative factor on w_t.*

Proof. Write

$$x(\theta_A, \theta_B, \phi) \;=\; x_A(\theta_A) \;+\; x_B(\theta_B) \;+\; x_{AB}(\theta_A, \theta_B) \;+\; x_{other}(\theta_A, \theta_B, \phi) \tag{25}$$

Then for head A we have

$$w_t^{(A)}(\theta_A) \;=\; \frac{1}{|y_t|\,|x(\theta_A, \theta_B, \phi)|} w_t^{*(A)}(\theta_A) \tag{26}$$

where w_t^* are the attention scores from the `No-Norm` case, which requires $x_A(\theta_A)$ and $x_B(\theta_B)$ to be linearly independent. Now we additionally require $|x(\theta_A, \theta_B, \phi)| \perp \theta_B, \phi$, with

$$|x| \;=\; \sqrt{|x_A|^2 \;+\; |x_B \;+\; x_{AB} \;+\; x_{other}|^2 \;+\; 2x_A^T(x_B \;+\; x_{AB} \;+\; x_{other})} \tag{27}$$

where we suppress parameter dependence for readability. Since $\sqrt{\cdot}$ is a monotonic function, this can only be satisfied if

$$|x_A|^2 \;+\; |x_B \;+\; x_{AB} \;+\; x_{other}|^2 \;+\; 2x_A^T(x_B \;+\; x_{AB} \;+\; x_{other}) \;\perp\; \theta_B, \phi \tag{28}$$

Repeating this process for head B gives

$$|x_B|^2 \;+\; |x_A \;+\; x_{AB} \;+\; x_{other}|^2 \;+\; 2x_B^T(x_A \;+\; x_{AB} \;+\; x_{other}) \;\perp\; \theta_A, \phi \tag{29}$$

Combining and collecting dependencies, we then have

$$|x_A|^2 \;=\; const \;\;\forall\;\; \theta_A \tag{30}$$

$$|x_B|^2 \;=\; const \;\;\forall\;\; \theta_B \tag{31}$$

$$(x_{AB} \;+\; 2x_A \;+\; 2x_B)^T x_{AB} \;+\; 2x_A^T x_B \;=\; const \;\;\forall\;\; \theta_A, \theta_B \tag{32}$$

$$(x_{other} + 2x_A + 2x_B + 2x_{AB})^T x_{other} \;=\; const \;\;\forall\;\; \theta_A, \theta_B, \phi \tag{33}$$

We can go one step further, noticing that each individual term carries a different functional dependence, and so must independently be constant[a]. We then have $\forall\;\; \mu, \nu \in \{A, B, AB, other\}$

$$|x_\mu| = const \quad \text{and} \quad x_\mu^T x_\nu = const \tag{34}$$

The requirements $|x_A(\theta_A)| = const\;\forall\;\theta_A$ and $|x_B(\theta_B)| = const\;\forall\;\theta_B$ mean that the semantic subspaces have a spherical structure defined by the L_2-norm $|\cdot|$.

Now consider the requirement $x_A(\theta_A)^T x_B(\theta_B) = const$. Say that θ_A and θ_B have N_A and N_B degrees of freedom, meaning that x_A and x_B have $N_A - 1$ and $N_B - 1$ respectively, since they each lose one by confinement to the sphere. Say that the constant is nonzero such that $x_A^T x_B \neq 0$. This means that there must be some direction i for which $x_{Ai} x_{Bi} \neq 0$. If we know all $N_A - 1$ coordinates of x_A, and all $N_B - 2$ coordinates of x_B except for direction i, then we also know the value of x_{Bi}, because it is fixed by the constant. However, this would mean that x_A and x_B are not independent, violating the condition $\theta_A \perp \theta_B$. The only way to satisfy independence is if $x_{Ai} x_{Bi} = 0\;\forall\;i$, ensuring that degrees of freedom on x_A and x_B never become entangled. Therefore, to satisfy semantic independence, we must have $x_A(\theta_A)^T x_B(\theta_B) = 0\;\forall\;\theta_A, \theta_B$. This means that the subspaces are not just linearly independent, but orthogonal.

We have shown the proof for semantic subspaces of x. As for Theorem 1, the same structure must be true for y_t by symmetry.

[a] N.B. If $|x_{AB}|^2 \propto x_A^T x_B$ then $|x_{AB}|^2 = const$ reduces to $x_A^T x_B = const$, which is already required.

Theorem 3. *`QKV-Norm`: Semantic subspaces must be linearly separable, reproducing the `No-Norm` case.*

Proof. We have

$$w_t^{(A)}(\theta_A) \;=\; \frac{1}{\left|k_t^{(A)}\right| \left|q^{(A)}\right|} w_t^{*(A)}(\theta_A) \tag{35}$$

where w_t^* are the attention scores from the `No-Norm` case, which requires $x_A(\theta_A)$ and $x_B(\theta_B)$ to be linearly independent. Use

$$x(\theta_A, \theta_B, \phi) \;=\; x_A(\theta_A) \;+\; x_B(\theta_B) \;+\; x_{other}(\theta_A, \theta_B, \phi) \tag{36}$$

and

$$\begin{aligned} q^{(A)}(\theta_A) \;&=\; W_Q^{(A)} x(\theta_A, \theta_B, \phi) \\ &=\; W_Q^{(A)} x_A(\theta_A) \;+\; W_Q^{(A)} x_B(\theta_B) \;+\; W_Q^{(A)} x_{other}(\theta_A, \theta_B, \phi) \end{aligned} \tag{37}$$

Since we already have the condition of linearly independent x_A, x_B, there must exist a linear projection operator P_A such that $P_A x_A = x_A$. Defining $W_Q^{(A)} = P_A$, we then have

$$q^{(A)}(\theta_A) \;=\; W_Q^{(A)} x_A(\theta_A) \tag{38}$$

This demonstrates that it is possible to separate linearly independent semantic subspaces on x. By symmetry of $w_t^{(A)}(\theta_A)$, the same must be true for y_t.

Theorem 4. Consider independent infinitesimal perturbations on queries $\epsilon^q \in \mathbb{R}^{N_{qkv}}$, keys $\epsilon_t^k \in \mathbb{R}^{N_{qkv}}$, and messages $\epsilon_t^m \in \mathbb{R}^{N_{qkv}}$. These propagate onto $\Delta x = \sum_t a_t m_t$ as

$$\epsilon^{\Delta x(q)} \xrightarrow[\epsilon^q \to 0]{\text{perturb q}} \mathop{\mathbb{E}}_{a_t}\left[m_t \tilde{k}_t^T\right] \epsilon^q \qquad \tilde{k}_t \;\triangleq\; k_t \;-\mathop{\mathbb{E}}_{a_t}\left[k_t\right] \tag{39}$$

$$\epsilon^{\Delta x(k)} \xrightarrow[\epsilon_t^k \to 0]{\text{perturb k}} \mathop{\mathbb{E}}_{a_t}\left[\tilde{m}_t \epsilon_t^{k\,T}\right] q \qquad \tilde{m}_t \;\triangleq\; m_t \;-\mathop{\mathbb{E}}_{a_t}\left[m_t\right] \tag{40}$$

$$\epsilon^{\Delta x(m)} \xrightarrow[\epsilon_t^m \to 0]{\text{perturb m}} \mathop{\mathbb{E}}_{a_t}\left[\epsilon_t^m\right] \tag{41}$$

where $\tilde{z}_t$ is the value of z_t measured from the attention-weighted centroid $\mathbb{E}_{a_t}[z_t] = \sum_t a_t z_t$.

Proof. Consider $q \to q + \epsilon^q$ where ϵ^q are infinitesimal perturbations on q. Then $\Delta x \to \Delta x + \epsilon^{\Delta x(q)}$ where by Taylor expansion we find

$$\epsilon^{\Delta x(q)} \;=\; \frac{\partial \Delta x}{\partial q} \epsilon^q \;+\; \mathcal{O}\left(\epsilon^{q2}\right) \tag{42}$$

where the leading term is a matrix $\frac{\partial \Delta x}{\partial q}$ acting on a vector ϵ^q. Differentiating gives

$$\frac{\partial \Delta x}{\partial q} \;=\; \sum_{ij} m_i \frac{\partial a_i}{\partial w_j} \frac{\partial w_j}{\partial q} \tag{43}$$

with $a_i = \texttt{softmax}_i(w_i)$ and $w_i = k_i^T q$, and we are using i, j, k etc to index over tokens instead of t, t', t'' etc, because this is more readable when we have many summations. Then

[continued in next box...]

[...continuing from previous box]

$$
\begin{aligned}
\frac{\partial a_i}{\partial w_j} &= \frac{\partial}{\partial w_j} \frac{e^{w_i}}{\sum_k e^{w_k}} \\
&= \frac{\delta_{ij} e^{w_i}}{\sum_k e^{w_k}} + e^{w_i} \left(- \frac{e^{w_j}}{\left(\sum_k e^{w_k}\right)^2} \right) \\
&= \frac{e^{w_i}}{\sum_k e^{w_k}} \left(1 - \frac{e^{w_j}}{\sum_l e^{w_l}} \right) \\
&= a_i \left(\delta_{ij} - a_j \right)
\end{aligned} \tag{44}
$$

and $\frac{\partial w_i}{\partial q} = k_i^T$, where we retain the transpose to indicate that this is an element of the dual vector space (i.e. covector). Inserting these results into our expression for $\epsilon^{\Delta x(q)}$ gives

$$
\begin{aligned}
\epsilon^{\Delta x(q)} &= \sum_{ij} m_i a_i \left(\delta_{ij} - a_j \right) k_j^T \epsilon^q \\
&= \sum_i m_i a_i \left(k_i - \sum_j a_j k_j \right)^T \epsilon^q \\
&= \sum_i m_i a_i \tilde{k}_i^T \epsilon^q \\
&= \mathop{\mathbb{E}}_{a_i} \left[m_i \tilde{k}_i^T \right] \epsilon^q
\end{aligned} \tag{45}
$$

This is the result for Eq. 39. Repeating the process for perturbations on k_i, we have

$$
\epsilon^{\Delta x(k)} = \sum_i \frac{\partial \Delta x}{\partial k_i} \epsilon_i^k + \mathcal{O}\left({\epsilon^k}^2 \right) \tag{46}
$$

and

$$
\begin{aligned}
\frac{\partial \Delta x}{\partial k_i} &= \sum_{jk} m_j \frac{\partial a_j}{\partial w_k} \frac{\partial w_k}{\partial k_i} \\
&= \sum_{jk} m_j a_j \left(\delta_{jk} - a_k \right) \delta_{ki} q^T \\
&= \sum_j m_j a_j \left(\delta_{ji} - a_i \right) q^T \\
&= a_i \tilde{m}_i q^T
\end{aligned} \tag{47}
$$

Therefore

$$
\epsilon^{\Delta x(k)} = \sum_i a_i \tilde{m}_i q^T \epsilon_i^k = \mathop{\mathbb{E}}_{a_i} \left[\tilde{m}_i {\epsilon_i^k}^T \right] q \tag{48}
$$

which is the result for Eq. 40. Finally,

$$
\begin{aligned}
\epsilon^{\Delta x(m)} &= \sum_i \frac{\partial \Delta x}{\partial m_i} \epsilon_i^m \\
&= \sum_i a_i \epsilon_i^m \\
&= \mathop{\mathbb{E}}_{a_i} \left[\epsilon_i^m \right]
\end{aligned} \tag{49}
$$

using $\frac{\partial \Delta x}{\partial m_i} = \frac{\partial}{\partial m_i} \sum_j a_j m_j = \sum_j a_j \delta_{ij} = a_i$. This is the result for Eq. 41.